Overview

Relevant source files

GDSTK (GDSII Tool Kit) is a C++ library with Python bindings for creating, manipulating, and writing GDSII and OASIS files. These file formats are standard in the semiconductor industry and other fields requiring precise geometric layout design. This page provides a high-level overview of GDSTK's purpose, architecture, and capabilities.

For details on specific architecture components, see Architecture. For usage examples, see Usage Examples.

Purpose and Scope

GDSTK serves as a comprehensive toolkit for programmatic creation and manipulation of complex integrated circuit layouts and similar geometric designs. It is designed as a successor to `Gdspy` with significant performance improvements and expanded capabilities.

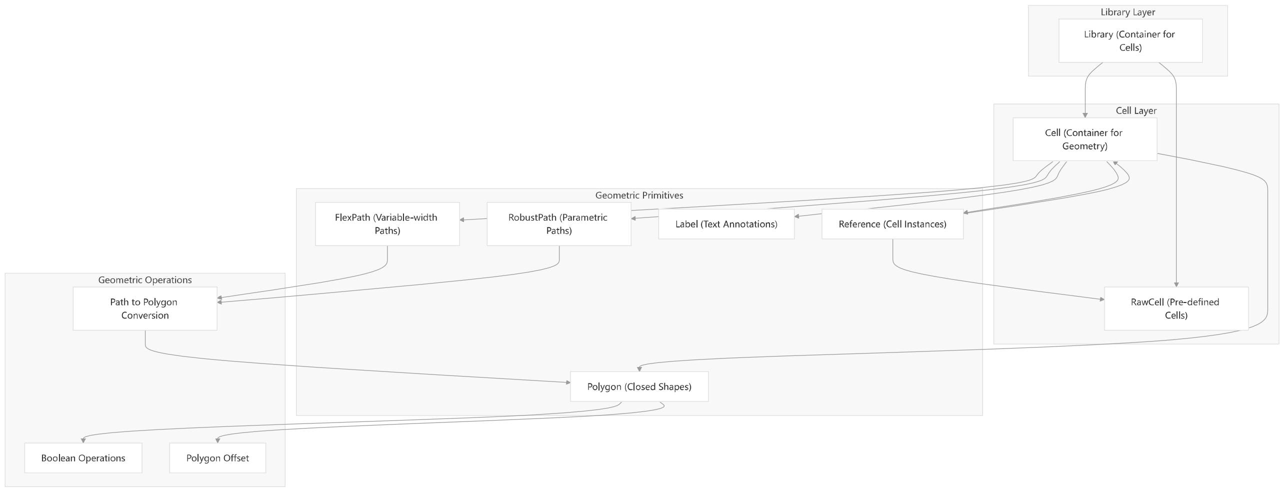
Key applications include:

- Electronic chip design
- Photonic integrated circuit layout
- MEMS (Micro-Electro-Mechanical Systems) design
- General-purpose geometric layout for fabrication

Sources: `README.md` 9-18

Core Architecture

GDSTK follows a hierarchical structure centered around geometric objects organized in cells, which are then collected in libraries.



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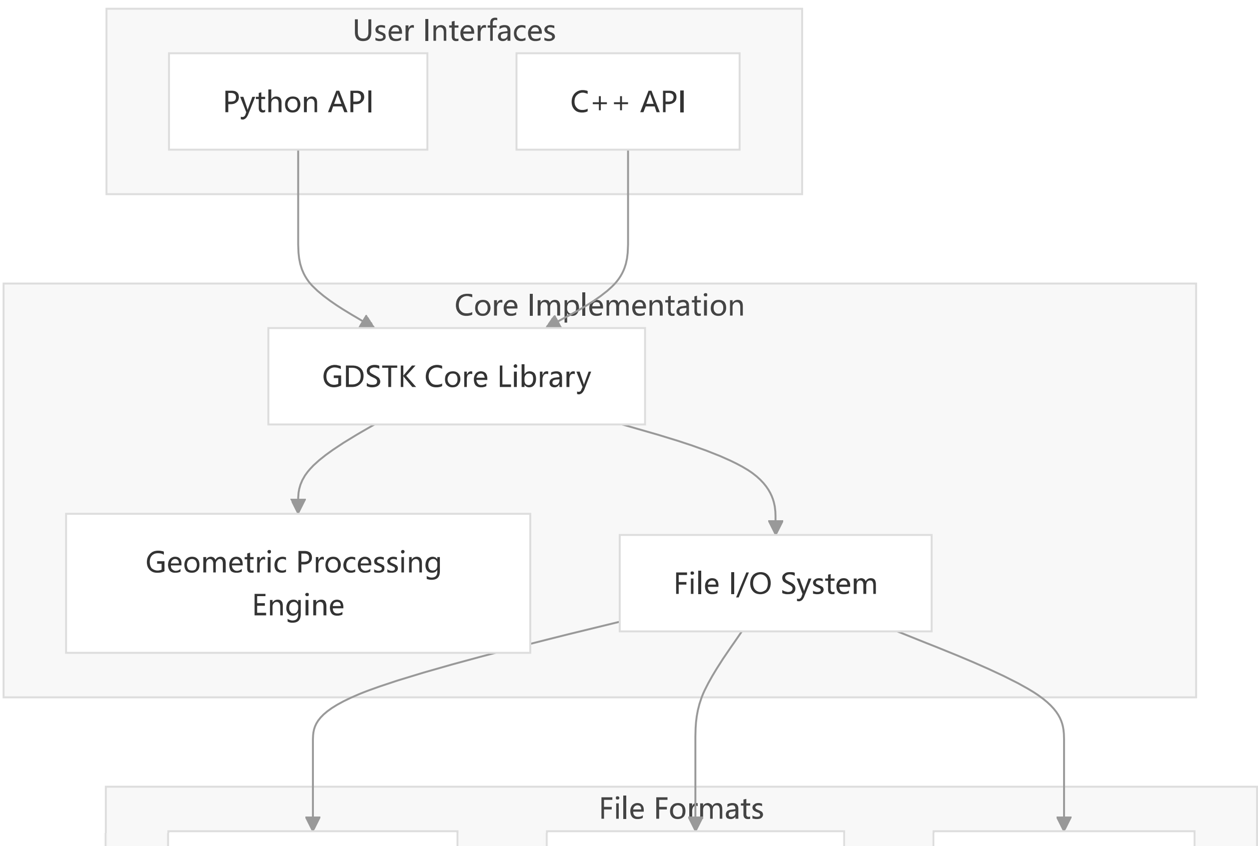
System Implementation

GDSTK is implemented as a C++ library with Python bindings, allowing users to work in either language while maintaining high performance.



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Key Features

Feature Category	Capabilities
Geometric Primitives	Polygons, FlexPaths, RobustPaths, Labels, References
Boolean Operations	Union, Intersection, Difference, XOR
Transformations	Translation, Rotation, Scaling, Reflection
Path Operations	Variable-width paths, Parametric paths, Path-to-polygon conversion
File Format Support	GDSII, OASIS, SVG export
Organization	Hierarchical cell structure, Cell references, Arrays
Optimization	Efficient boolean operations, Fast point-in-polygon testing

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Performance

GDSTK is designed for high performance, with significant speed improvements over its predecessor Gdspy. The C++ implementation delivers exceptional performance for complex geometric operations.

Benchmark	Speed Improvement vs. Gdspy 1.6.13
Creating 10k rectangles	16.5× faster
Bounding box calculation	216× faster
FlexPath operations	178× faster
RobustPath operations	19.7× faster
Boolean operations	4.19× faster
GDS file reading	28.5× faster

Memory usage is also significantly reduced, with most objects requiring 24-71% less memory

Development History

GDSTK is currently at version 0.9.60 (as of April 2025) and has been under active development since its initial release in October 2020. It continues to receive regular updates with new features, performance improvements, and bug fixes.

Recent additions include:

- Support for 32-bit layers and datatypes
- Support for Python 3.13
- Various bug fixes and performance improvements

Sources: `CHANGELOG.md` 3-38 `include/gdstk/gdstk.hpp` 14 `pyproject.toml` 10

System Requirements

GDSTK has the following dependencies:

- For C++ library:
  - zlib
  - qhull
- Additional dependencies for Python module:
  - Python 3.9+
  - NumPy

Sources: `README.md` 44-53 `pyproject.toml` 30-34

License

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